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Theoretical Physics Archive
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Date: October 25, 2026
To: The Editorial Board / Chief Editor
Journal Target: *Journal of Mathematical Physics* / *Journal of Geometry and Physics*
Subject: Submission of Original Research Manuscript

Dear Editor,

I am pleased to submit my original research manuscript titled **"Oscillating Dimensional Trajectories on E_8 Lattices & Gravitational Modification"** for consideration as a peer-reviewed research article in your esteemed journal.

This manuscript establishes a novel mathematical framework unifying quantum subatomic mechanics, cosmological scale dynamics, and gravitational field modification. Rather than relying on static compactified internal dimensions, the paper introduces an active 1D–8D dynamic dimensional trajectory operator, $D(n) = 8 - |(n \bmod 14) - 7|$, defined across the 240 root vectors of the exceptional Lie algebra E_8 lattice. Modulated by golden-ratio scaling factors $\varphi^{(-1)^n \cdot D(n)}$, the model demonstrates how quantum wave-function collapse, cosmic inflationary expansion, and localized metric cancellation (anti-gravity / inertial decoupling) emerge naturally as dual phases of a unified recursive feedback loop.

> **Key Contributions & Theoretical Scope:**
> • **Mathematical Unification:** Continuous recursive mapping between 8D E_8 Lie algebra root projections, 4D Mandelbrot iterative geometry, and golden-ratio scaling invariants.
> • **Gravitational Modification:** Derivation of effective negative energy density states $\rho_{eff} < 0$ and metric phase slipping for inertial decoupling.
> • **Mathematics Subject Classification (MSC):** 81T30 (String & Universal Lattices), 51M20 (Polyhedra & Polytopes), 83F05 (Relativistic Cosmology).

I confirm that this manuscript represents my original work, has not been published previously, and is not currently under consideration for publication elsewhere. I select the **traditional subscription-based publishing route** (with zero Author Processing Charges) and declare no competing financial or institutional conflicts of interest.

Full computational models, interactive geometric visualizations, and supporting technical assets are archived publicly at **donniebugden.com**.

Thank you for your time, consideration, and evaluation of this work.

Sincerely,

Donald Bugden

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